

Unsupervised learning

Unsupervised learning

- Unsupervised learning, also known as unsupervised machine learning, uses machine learning algorithms to analyze and cluster unlabelled datasets.

ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSize	BareNuc	BlandChrom	NormNucl	Mit	Class
1000025	5	1	1	1	2	1	3	1	1	benign
1002945	5	4	4	5	7	10	3	2	1	benign
1015425	3	1	1	1	2	2	3	1	1	malignant
1016277	6	8	8	1	3	4	3	7	1	benign
1017023	4	1	1	3	2	1	3	1	1	benign
1017122	8	10	10	8	7	10		7	1	malignant
1018099	1	1	1	1	2	10	3	1	1	benign
1018561	2	1	2	H	2	1	3	1	1	benign
1033078	2	1	1	1	2	1	1	1	5	benign
1033078	4	2	1	1	2	1	2	1	1	benign

labels

Customer Id	Age	Edu	Years Employed	Income	Card Debt	Other Debt	Address	DebtIncomeRatio
1	41	2	6	19	0.124	1.073	NBA001	6.3
2	47	1	26	100	4.582	8.218	NBA021	12.8
3	33	2	10	57	6.111	5.802	NBA013	20.9
4	29	2	4	19	0.681	0.516	NBA009	6.3
5	47	1	31	253	9.308	8.908	NBA008	7.2
6	40	1	23	81	0.998	7.831	NBA016	10.9
7	38	2	4	56	0.442	0.454	NBA013	1.6
8	42	3	0	64	0.279	3.945	NBA009	6.6
9	26	1	5	18	0.575	2.215	NBA006	15.5
10	47	3	23	115	0.653	3.947	NBA011	4
11	44	3	8	88	0.285	5.083	NBA010	6.1
12	34	2	9	40	0.374	0.266	NBA003	1.6

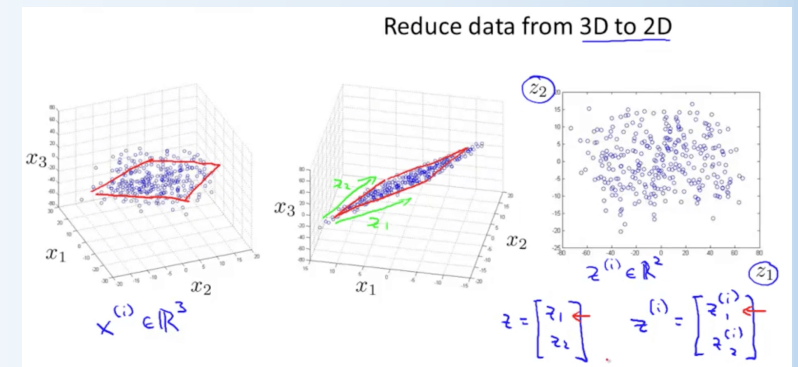
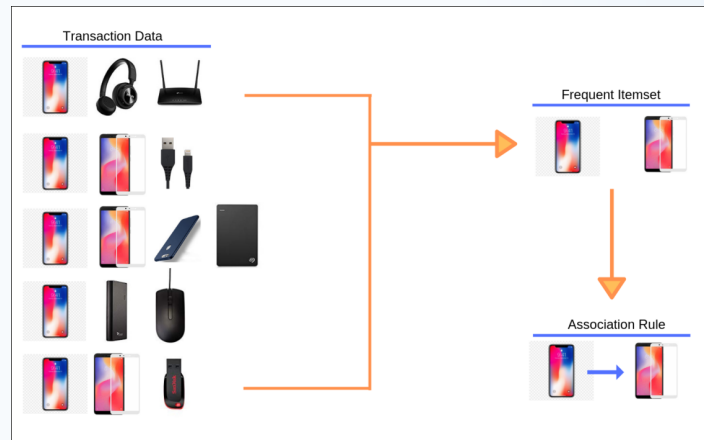
unlabeled

- These algorithms discover hidden patterns or data groupings without the need for human intervention.
- Its ability to discover similarities and differences in information make it the ideal solution for exploratory data analysis, cross-selling strategies, customer segmentation, and image recognition.

Common unsupervised learning approaches

Unsupervised learning models are utilized for three main tasks—

- Clustering
- Association
- Dimensionality reduction



Clustering

Clustering

- Clustering is a data mining technique which groups unlabelled data based on their similarities or differences. Clustering algorithms are used to process raw, unclassified data objects into groups represented by structures or patterns in the information.
- Clustering algorithms can be categorized into a few types, specifically exclusive, overlapping, hierarchical, and probabilistic.

Exclusive and Overlapping Clustering

- Exclusive clustering is a form of grouping that stipulates a data point can exist only in one cluster.
- This can also be referred to as “hard” clustering. The K-means clustering algorithm is an example of exclusive clustering.

Clustering

K-means clustering

- It is a common example of an exclusive clustering method where data points are assigned into K groups, where K represents the number of clusters based on the distance from each group's centroid.
- The data points closest to a given centroid will be clustered under the same category.
- A larger K value will be indicative of smaller groupings with more granularity whereas a smaller K value will have larger groupings and less granularity.
- K-means clustering is commonly used in market segmentation, document clustering, image segmentation, and image compression.

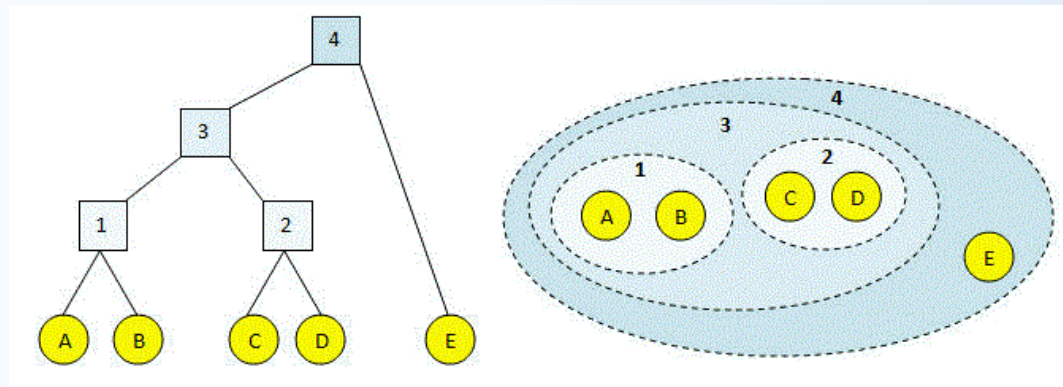
Overlapping clusters differs from exclusive clustering in that it allows data points to belong to multiple clusters with separate degrees of membership.

“Soft” or fuzzy k-means clustering is an example of overlapping clustering.

Clustering

Hierarchical clustering

- Hierarchical clustering, also known as hierarchical cluster analysis (HCA), is an unsupervised clustering algorithm that can be categorized in two ways; they can be agglomerative or divisive.
- Agglomerative clustering is considered a “bottoms-up approach.” Its data points are isolated as separate groupings initially, and then they are merged together iteratively on the basis of similarity until one cluster has been achieved.



Association Rules

Association Rules

- An association rule is a rule-based method for finding relationships between variables in a given dataset.
- These methods are frequently used for market basket analysis, allowing companies to better understand relationships between different products.
- Understanding consumption habits of customers enables businesses to develop better cross-selling strategies and recommendation engines.
- Examples of this can be seen in Amazon's "Customers Who Bought This Item Also Bought" or Spotify's "Discover Weekly" playlist. While there are a few different algorithms used to generate association rules, such as Apriori, Eclat, and FP-Growth, the Apriori algorithm is most widely used.

Dimensionality Reduction

Dimensionality reduction

- While more data generally yields more accurate results, it can also impact the performance of machine learning algorithms (e.g. overfitting) and it can also make it difficult to visualize datasets.
- Dimensionality reduction is a technique used when the number of features, or dimensions, in a given dataset is too high.
- It reduces the number of data inputs to a manageable size while also preserving the integrity of the dataset as much as possible.
- It is commonly used in the preprocessing data stage.

Dimensionality Reduction

Principal component analysis

- Principal component analysis (PCA) is a type of dimensionality reduction algorithm which is used to reduce redundancies and to compress datasets through feature extraction.
- This method uses a linear transformation to create a new data representation, yielding a set of "principal components." The first principal component is the direction which maximizes the variance of the dataset.
- While the second principal component also finds the maximum variance in the data, it is completely uncorrelated to the first principal component, yielding a direction that is perpendicular, or orthogonal, to the first component.
- This process repeats based on the number of dimensions, where a next principal component is the direction orthogonal to the prior components with the most variance.

K Means Clustering

K-Means Algorithm

K-Means Clustering Algorithm

- K-Means Clustering is an unsupervised learning algorithm that is used to solve the clustering problems in machine learning or data science.
- In K-Means algorithm we group the unlabelled dataset into different clusters.
- Here K defines the number of pre-defined clusters that need to be created in the process, as if $K=2$, there will be two clusters, and for $K=3$, there will be three clusters, and so on.
- It is an iterative algorithm that divides the unlabelled dataset into k different clusters in such a way that each dataset belongs only one group that has similar properties.
- It is a centroid-based algorithm, where each cluster is associated with a centroid. The main aim of this algorithm is to minimize the sum of distances between the data point and their corresponding clusters.

K-Means Algorithm

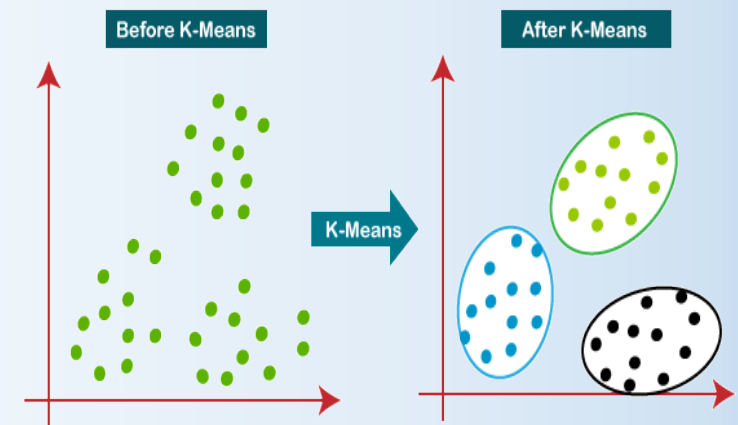
K-Means Clustering Algorithm

- The algorithm takes the unlabelled dataset as input, divides the dataset into k-number of clusters, and repeats the process until it does not find the best clusters. The value of k should be predetermined in this algorithm.

The k-means clustering algorithm mainly performs two tasks:

- Determines the best value for K center points or centroids by an iterative process.
- Assigns each data point to its closest k-center. Those data points which are near to the particular k-center, create a cluster.

Hence each cluster has datapoints with some commonalities, and it is away from other clusters.



Working of K-Means Algorithm

The working of the K-Means algorithm is explained in the below steps:

Step-1: Select the number K to decide the number of clusters.

Step-2: Select random K points or centroids. (It can be other from the input dataset).

Step-3: Assign each data point to their closest centroid, which will form the predefined K clusters.

Step-4: Calculate the variance and place a new centroid of each cluster.

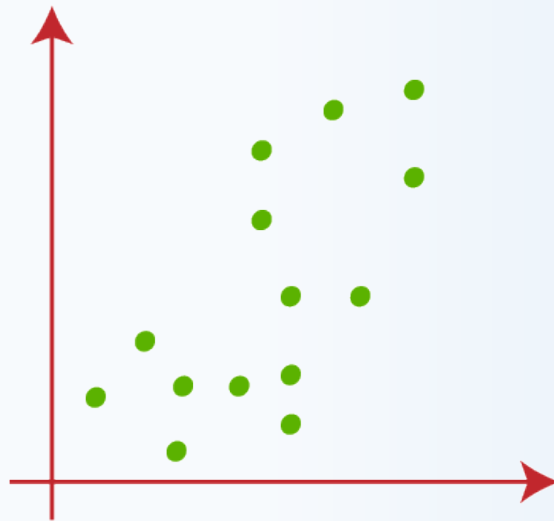
Step-5: Repeat the third steps, which means reassign each datapoint to the new closest centroid of each cluster.

Step-6: If any reassignment occurs, then go to step-4 else go to finish.

Step-7: The model is ready.

Working of K-Means Algorithm with Example

Suppose we have two variables M1 and M2. The x-y axis scatter plot of these two variables is given below:



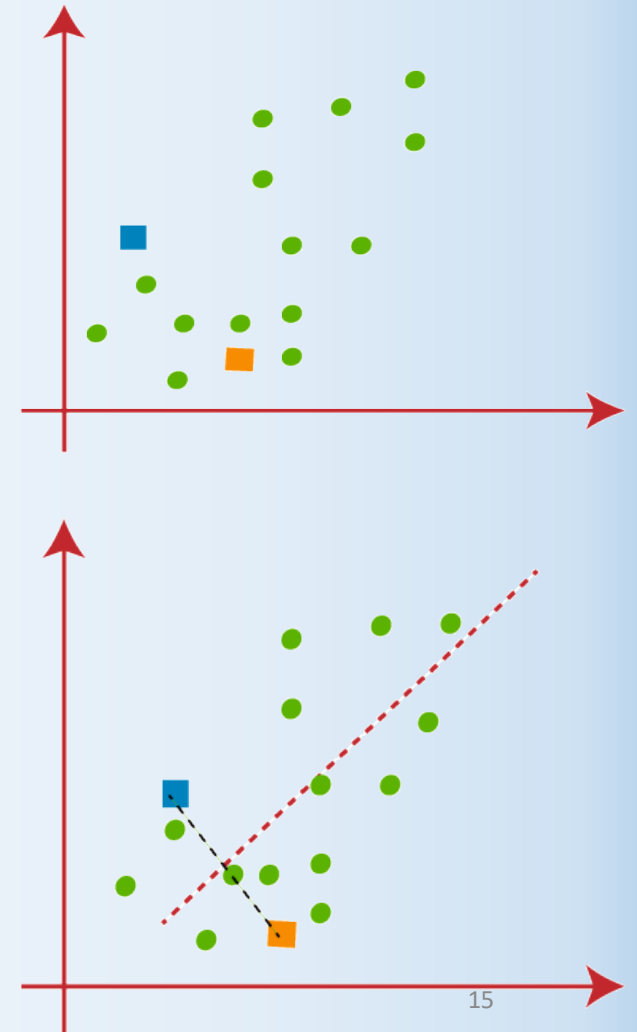
Let's take number k of clusters, i.e., $K=2$, to identify the dataset and to put them into different clusters. It means here we will try to group these datasets into two different clusters.

Working of K-Means Algorithm with Example

We need to choose some random k points or centroid to form the cluster. These points can be either the points from the dataset or any other point. So, here we are selecting the two points as k points, which are not the part of our dataset.

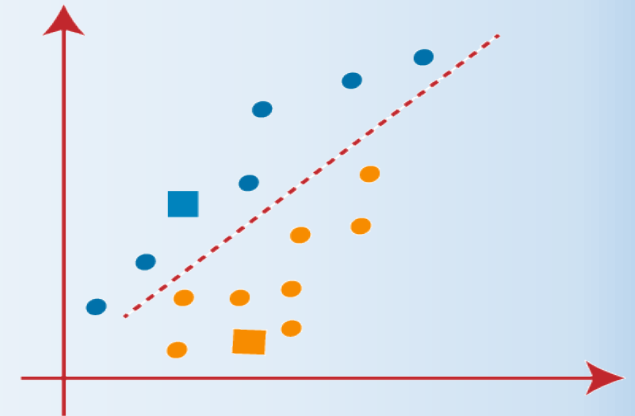
Now we will assign each data point of the scatter plot to its closest K -point or centroid. We will compute it by applying some mathematics that we have studied to calculate the distance between two points.

So, we will draw a median between both the centroids.

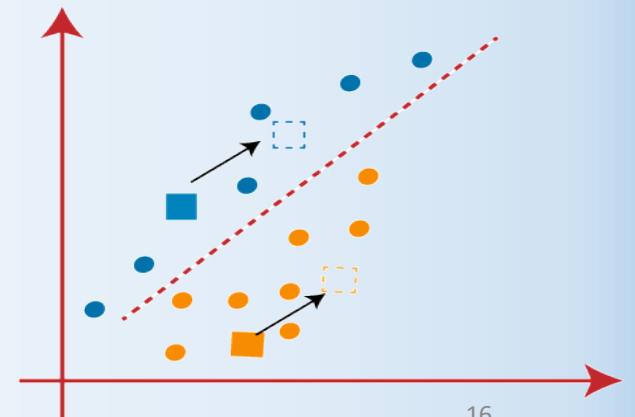


Working of K-Means Algorithm with Example

From the above steps, it is clear that points left side of the line is near to the K1 or blue centroid, and points to the right of the line are close to the yellow centroid. Let's color them as blue and yellow for clear visualization.



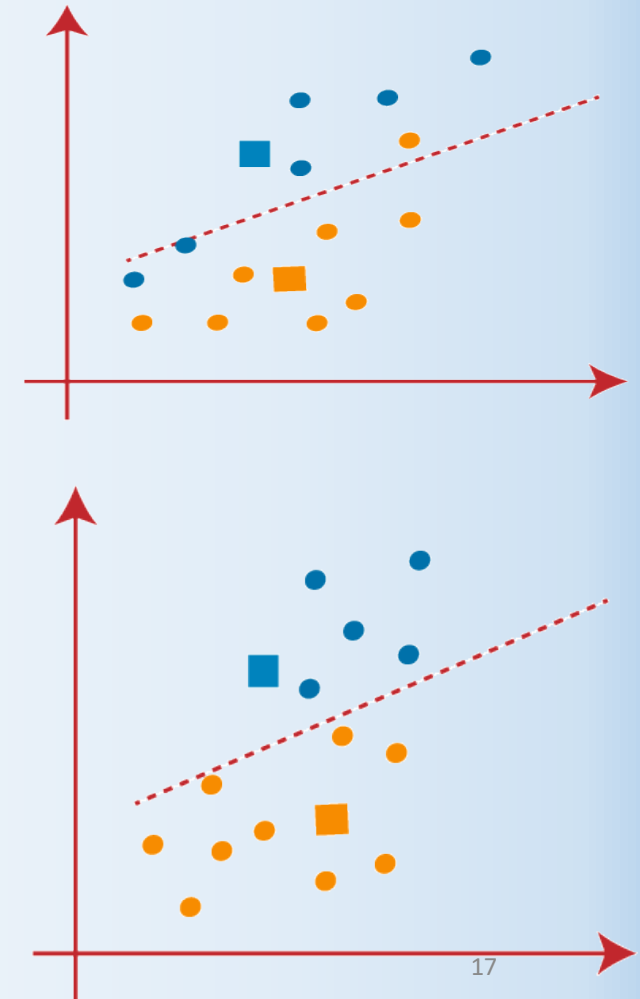
As we need to find the closest cluster, so we will repeat the process by choosing a new centroid. To choose the new centroids, we will compute the center of gravity of these centroids, and will find new centroids as below:



Working of K-Means Algorithm with Example

Next, we will reassign each datapoint to the new centroid. For this, we will repeat the same process of finding a median line. The median will be like adjacent image:

From the previous step, we can see, one yellow point is on the left side of the line, and two blue points are right to the line. So, these points will be assigned to new centroids.

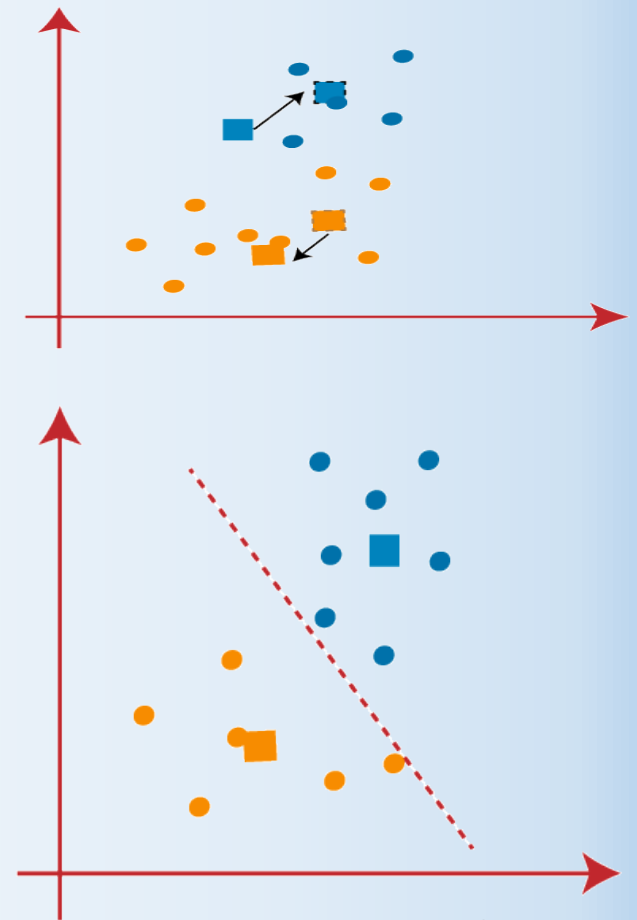


Working of K-Means Algorithm with Example

As reassignment has taken place, so we will again go to the step-4, which is finding new centroids or K-points.

We will repeat the process by finding the center of gravity of centroids, so the new centroids will be as shown in the adjacent image:

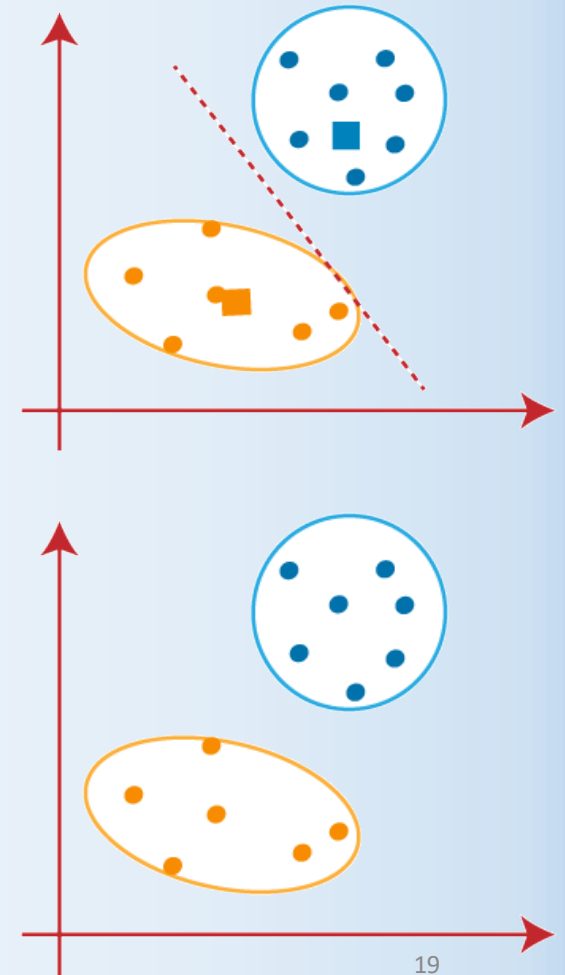
As we got the new centroids so again will draw the median line and reassign the data points. So, the image will be:



Working of K-Means Algorithm with Example

We can see in the above step; there are no dissimilar data points on either side of the line, which means our model is formed. Consider the adjacent image:

As our model is ready, so we can now remove the assumed centroids, and the two final clusters will be as shown in the adjacent image:



How to choose the value of K

The performance of the K-means clustering algorithm depends upon highly efficient clusters that it forms. But choosing the optimal number of clusters is a big task. There are some different ways to find the optimal number of clusters.

Elbow Method

The Elbow method is one of the most popular ways to find the optimal number of clusters. This method uses the concept of WCSS value.

WCSS stands for **Within Cluster Sum of Squares**, which defines the total variations within a cluster.

To measure the distance between data points and centroid, we can use any method such as Euclidean distance or Manhattan distance.

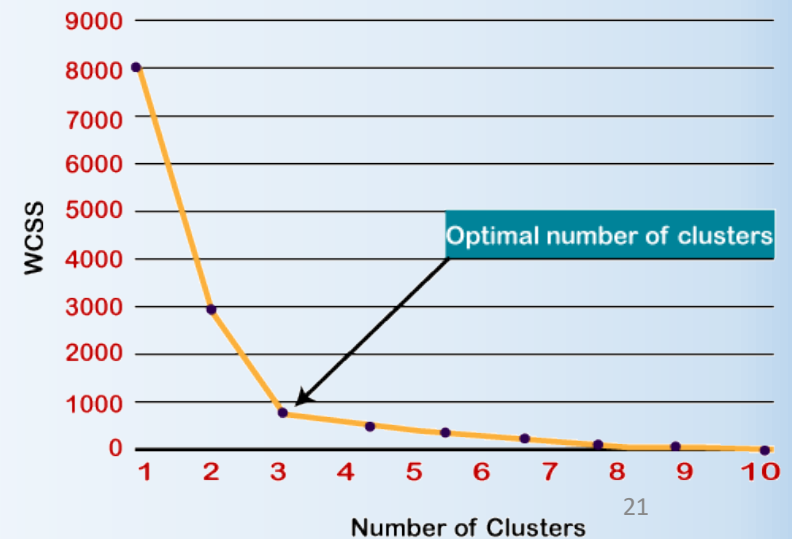
Steps to choose the value of K

To find the optimal value of clusters, the elbow method follows the below steps:

- It executes the K-means clustering on a given dataset for different K values (ranges from 1-10).
- For each value of K, calculates the WCSS value.
- Plots a curve between calculated WCSS values and the number of clusters K.
- The sharp point of bend or a point of the plot looks like an arm, then that point is considered as the best value of K.

Since the graph shows the sharp bend, which looks like an elbow, hence it is known as the elbow method.

The graph for the elbow method looks like the adjacent image:



Important Hyperparameters

n_clusters : int, default=8

The number of clusters to form as well as the number of centroids to generate.

Init {'k-means++', 'random'}

'k-means++' : selects initial cluster centroids using sampling based on an empirical probability distribution of the points' contribution to the overall inertia.

'random': choose n_clusters observations (rows) at random from data for the initial centroids.

n_init : int, default=10

Number of time the k-means algorithm will be run with different centroid seeds.

Application of Clustering

Clustering is widely used in many industries. Below are some commonly known applications of clustering technique in Machine Learning:

- **In Identification of Cancer Cells:** The clustering algorithms are widely used for the identification of cancerous cells. It divides the cancerous and non-cancerous data sets into different groups.
- **In Search Engines:** Search engines also work on the clustering technique. The search result appears based on the closest object to the search query. It does it by grouping similar data objects in one group that is far from the other dissimilar objects. The accurate result of a query depends on the quality of the clustering algorithm used.
- **Customer Segmentation:** It is used in market research to segment the customers based on their choice and preferences.
- **In Biology:** It is used in the biology stream to classify different species of plants and animals using the image recognition technique.
- **In Land Use:** The clustering technique is used in identifying the area of similar lands use in the GIS database. This can be very useful to find that for what purpose the particular land should be used, that means for which purpose it is more suitable.

Principal Component Analysis

PCA and Dimensionality Reduction

Why PCA and why Dimensionality Reduction?

Imagine we have a large dataset, which has multiple dimensions or features. Just by looking at it and trying to perform EDA and analyzing the hidden patterns in the data is a tedious job.

It is difficult to try to understand hidden patterns just by looking at the data.

This is where the PCA comes to our rescue. The idea behind PCA is simply to find a low-dimension set of axes that summarize data.

Example

We have a dataset composed of a set of properties from Laptops. These properties describe each Laptop by its size, color, screen size, weight, processor, number of USB slots, O.S., Touch screen or no Touch screen, and so on. However, many of those features will **measure related properties** and thus are going to be redundant. Therefore, we should remove this redundancy and describe each laptop with fewer properties. This is exactly what PCA does.

For example, think about the screen size as a feature of Laptops, almost every example has a screen size almost similar, **hence we can conclude that this feature has a low variance** (On average, most popular laptops have screen sizes that range between 13 to 15 inches.), so this feature will make all laptops look the same, but they are pretty different from each other.

PCA and Dimensionality Reduction

Why PCA and why Dimensionality Reduction?

Do not think that PCA is just dropping the feature of the screen size of Laptops and thus by considering the processor feature and other features which has significant variance in it is reducing the dimensions of the given dataset.

PCA does not drop any feature present in the original dataset and arrives at Principal Components.

The Algorithm actually constructs a new set of properties which is a combination of the old ones. Mathematically PCA performs a

Linear Transformation moving the original set of features to a new space composed by the Principal

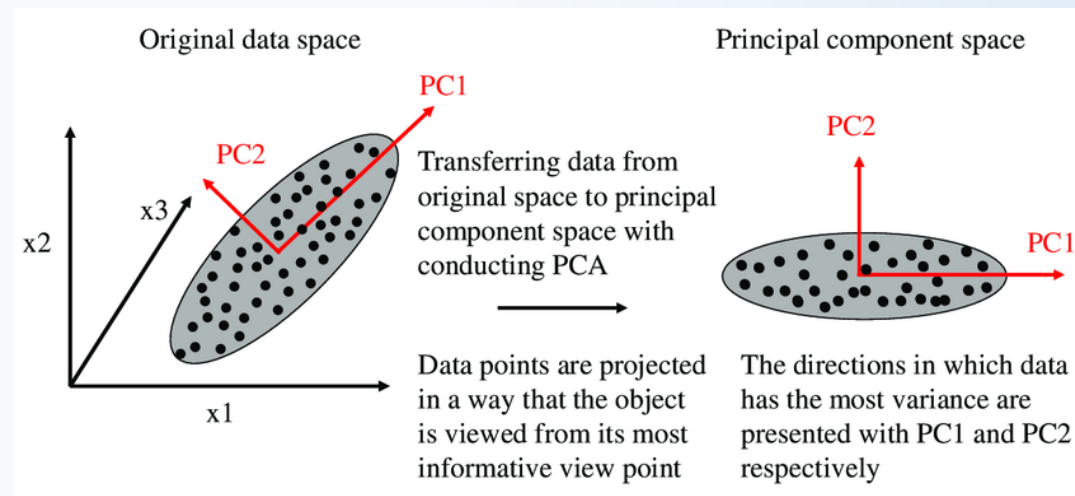
Component. These new features do not have any real meaning except algebraic, therefore do not think that by combining features, linearly you will find new features that you have never thought could exist.

PCA and Dimensionality Reduction

How does the new PCA space look like?

In the new feature space, we are trying to find some properties that strongly differ across the classes. As discussed in the above example, some properties that present low variance are not useful, it will make the laptops look the same. On the other hand, PCA looks for properties that show the maximum amount of variation across classes as possible to create the Principal Component Space. The algorithm uses the concepts of Variance Matrix, Covariance Matrix, Eigenvector, and EigenValues pairs to perform PCA, providing a set of eigenvectors and its respective eigenvalues as a result.

The concepts of Covariance Matrix, Eigenvalues, and Eigenvectors are explained in the section below on Mathematical concepts behind PCA.



Steps in PCA

The steps followed in PCA are as mentioned below:

1. Standardization of the continuous variables of the dataset.
2. Computing the Co-variance Matrix to identify Co-relations.
3. Computing the Eigen Values and Eigenvectors of the covariance Matrix to identify the Principal Components.
4. Deciding on the Principal Components to be kept for further analysis based on the variation in the Components using the Scree Plot.
5. Recast the data along the Principal Component's axes.

Steps in PCA

Step 1: Standardization

- The reason why it is important to perform standardization before PCA is that the latter is extremely sensitive regarding the variances of the variables. i.e. if there are large differences between the ranges of initial variables, those variables with larger ranges will dominate over those with small ranges(For example, a variable that ranges between 0 and 100 will dominate over a variable that ranges between 0 and 1), which can cause biased results.
- So transforming the data to comparable scales will prevent this problem.

Steps in PCA

Statistical Tests to be done for PCA

The following are the 2 tests that we perform on the dataset to identify whether to perform PCA on the given dataset or not to perform the same.

1. Bartlett's Test of Sphericity :

It tests the hypothesis that the variables are uncorrelated within the population.

H₀: Null Hypothesis: All variables in the data are uncorrelated.

H_a: Alternate Hypothesis: At least one pair of variables in the data are correlated
if the null hypothesis cannot be rejected, then PCA is not advisable.

If the p-value is small, then we can reject the Null Hypothesis and agree that there is at least one pair of variables in the data which are correlated hence PCA is recommended.

2. The Kaiser- Meyer- Olkin (KMO) – a Measure of Sampling Adequacy (MSA) is an index used to determine how appropriate PCA is. Generally, if **MSA is less than 0.5**, PCA is not recommended, since no reduction is expected.

On the other hand, **MSA > 0.7 is expected to provide a considerable reduction in the dimension** and extraction of meaningful components.

Steps in PCA

Step 2: Covariance Matrix Computation

In this step, we will try to identify if there is any relationship between the variables in the dataset. As sometimes, variables are highly correlated in a way such that the information contained in them is redundant. So as to identify these co-relations, we compute the Covariance Matrix.

STEP 3: Computing Eigenvectors and Eigenvalues

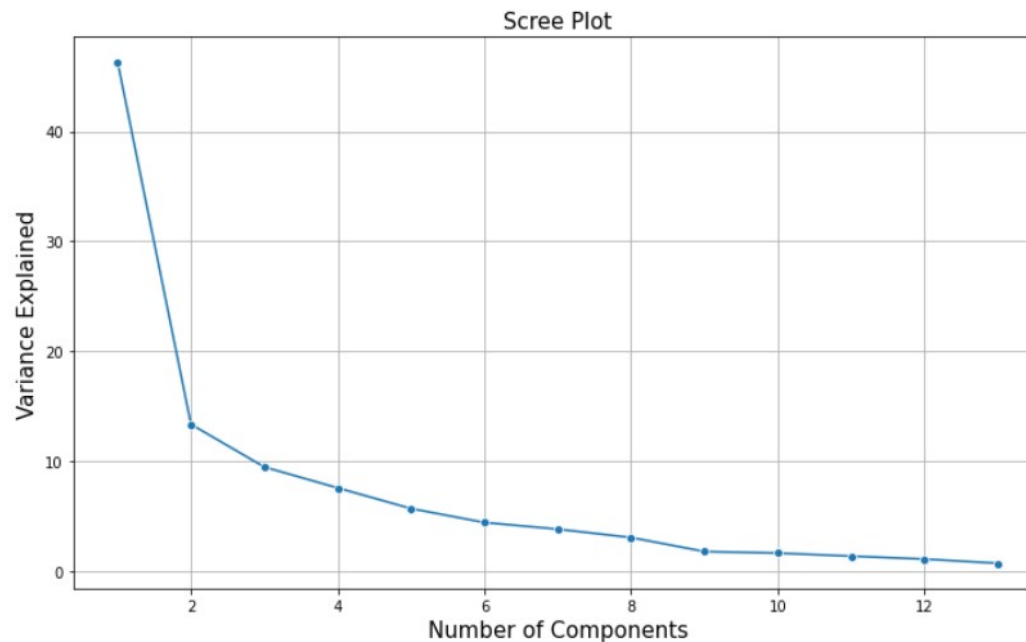
- Eigenvectors and Eigenvalues are the Mathematical Concepts of Linear Algebra that we've to calculate from the Covariance Matrix to find out the Principal Components of the dataset
- Principal Components represent the directions of the data that specify a maximum amount of variance i.e. the lines that capture most information present in the data.
- To put it in a simpler way, just consider Principal Components as new axes that provide the best angle to visualize and evaluate the data, in order that the difference between the observations is best visible.

Steps in PCA

Step 4: Scree Plot to decide on the Number of PCS

A Scree plot always displays the Eigenvalues in a downward curve, ordering the eigenvalues from largest to smallest. According to the Scree test, the "elbow" of the graph where the Eigenvalues seem to level off is found and factors or components to the left of this point should be retained as significant.

The Scree Plot is used to determine the number of Principal Components to keep in a Principal Component Analysis(PCA).



Steps in PCA

Step 5: Recasting the Data along with Principal Component Axes

In this step, we are going to use the feature vector formed using the Eigenvectors of the Covariance Matrix, to reorient the data from the original axes to the ones represented by the Principal Components. (Hence the name Principal Component Analysis).



Advantages and of PCA

Advantages of PCA

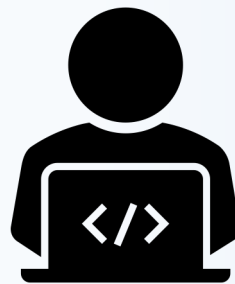
1. Removes correlated Features.
2. Improves Algorithm Performance
3. Reduces Overfitting
4. Improves visualization of the data.

Limitations of PCA

1) Model Performance: PCA can lead to a reduction in Model Performance on datasets with no or low feature co-relation or does not meet the assumptions of linearity.

2) Outliers: PCA is affected by outliers. Hence treating outliers is important before performing PCA.

3) Interpretability: After performing PCA on the dataset original features will turn into Principal Components which is a Linear Combination of the original features. Hence, these principal components are not as readable as the original features.



Keep Learning..... Keep Coding..... Keep going.....